

Label	Design Requirement Description
CONFIG_0	All registers must initialise to their defined reset values on active-high reset
CONFIG_1	All registers must store the maximum positive value (0xFFFF) when written
CONFIG_2	Registers must hold their value while reading
CONFIG_3	All registers must store the maximum negative value (0x8000) when written
CONFIG_4	All registers must store zero (0x0000) when written
CONFIG_5	All registers must correctly store any random 16-bit value when written
CONFIG_6	Write disabled must not modify any register regardless of data_in value

Stimulus Generation	Functional Coverage
Directed reset at start of simulation	-
Directed — iterate all 8 addresses with MAXPOS, write enabled	-
disavle write, iterate all 8 addresses after CONFIG_1 writes	-
Directed — iterate all 8 addresses with MAXNEG, write enabled	-
Directed — iterate all 8 addresses with ZERO, write enabled	-
Random — \$random data written to all 8 addresses sequentially	-
Directed — write disabled, MAXPOS presented on data_in, all addresses iterated	-

Functionality Check
Golden model comparison using reset_assoc array against data_out
Golden model comparison using expected_assoc
Golden model comparison — data_out must match expected_assoc
Golden model comparison using expected_assoc
Golden model comparison using expected_assoc
Golden model comparison using expected_assoc
Golden model comparison — data_out must remain equal to expected_assoc